

Linux Fundamentals: Notes

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1. The Linux Terminal

The Linux terminal is a text-based way to interact with a computer. You type a command, press **Enter**, and Linux performs an action or displays information.

Commands are usually entered one at a time. A command may show text, create or change something, or produce no visible output. No output does not always mean a command failed; some commands work silently.

2. Current Location: `pwd`

`pwd`

`pwd` means “print working directory.” It displays the folder you are currently working in.

Knowing your current location helps you understand where a new file or folder will be created and which files a command may affect.

3. Listing Files: `ls` and `ls -l`

`ls`

`ls` lists the files and folders in the current directory.

`ls -l`

The `-l` option shows a longer listing with additional information, including permissions, ownership, file size, and modification time.

Use `ls` to see what is in a folder. Use `ls -l` when you need more detail about its contents.

4. Creating a Directory: `mkdir`

`mkdir practice`

`mkdir` means “make directory.” This command creates a folder named `practice`.

Folders help organize related files. If a folder already exists, Linux may display a message instead of creating another one.

5. Moving Between Directories: `cd`

```
cd practice
```

`cd` means “change directory.” This command moves you into the `practice` folder.

After changing directories, use `pwd` to check your current location. The location matters because commands such as `touch` create files in the current directory.

6. Creating a File: `touch`

```
touch hello.txt
```

`touch` creates an empty file named `hello.txt` if it does not already exist.

If a file with that name already exists, `touch` updates its modification time. It does not normally erase the file’s contents.

7. Viewing File Contents: `cat`

```
cat hello.txt
```

`cat` displays the contents of a file in the terminal. If the file is empty, the command may show no text.

Use it for short files. For larger files, tools such as `less` are more convenient because they let you read the content page by page.

8. File Permissions

Linux permissions control what users can do with files and directories. The basic permission types are:

- **Read (r):** view a file’s contents or list a directory’s contents.
- **Write (w):** change a file or add and remove items in a directory.
- **Execute (x):** run a program or access a directory.

Permissions are shown for three categories:

- **Owner:** the user who owns the file.
- **Group:** users who belong to the file’s group.
- **Others:** everyone else.

Changing permissions with `chmod`

```
chmod 600 hello.txt
```

`chmod` changes permissions. The mode 600 gives the owner permission to read and write the file, while the group and others receive no permissions.

Only change permissions on files you own or are authorized to manage. A permission change can affect whether other users or programs can access a file.

Checking permissions

```
ls -l hello.txt
```

This displays the file's details, including its permissions. A permission string such as `-rw-----` means:

- The first `-` indicates a regular file.
- `rw-` means the owner can read and write it.
- The next `----` means the group has no permissions.
- The final `---` means other users have no permissions.

9. User Information: `whoami` and `id`

```
whoami
```

`whoami` displays the name of the user account currently being used.

```
id
```

`id` displays the current user's identification information and group memberships.

Linux uses accounts and groups to manage access. A user's group membership can affect which files and resources they can use. Avoid sharing account details if they are private.

10. Running Programs: `ps`

```
ps
```

`ps` displays information about processes running in the terminal's current context. A **process** is a program that is currently running.

The output can include a process ID and the command associated with that process. A process ID helps identify a running program.

11. Commands Practiced

Command	Meaning
pwd	Displays the current working directory.
ls	Lists files and directories.
mkdir practice	Creates a directory named practice.
cd practice	Moves into the practice directory.
touch hello.txt	Creates an empty file named hello.txt.
ls -l	Lists files with additional details.
cat hello.txt	Displays the file's contents.
chmod 600 hello.txt	Sets owner read/write permission and removes group and other permissions.
whoami	Displays the current user account.
id	Displays user and group information.
ps	Displays running processes.

12. Safe Linux Practice

- Check your current directory with `pwd` before creating or changing files.
- Practice with test files and folders you created yourself.
- Read a command before running it, especially if you found it online.
- Be cautious with commands that delete files or change permissions.
- Do not include passwords, access tokens, or private information in notes or screenshots.
- Ask your supervisor before using security tools or testing systems outside your own practice environment.